ALL SYSTEMS ONLINE. DIAGNOSTIC MODE: ACTIVE.

The Azure Storage Blueprint

Architectural Diagnostics & System Resolutions

Optimal Storage for Legacy Lift and Shift

You are migrating a legacy on-premises application to Azure. The application relies on native file system APIs to share data concurrently with other applications and requires a standard Server Message Block (SMB) protocol. Which Azure Storage data service is best suited for this migration?

A) Azure Blobs

B) Azure Disks

C) Azure Tables

D) Azure Files

E) Azure Queues

Optimal Storage for Legacy Lift and Shift

A) Azure Blobs

B) Azure Disks

C) Azure Tables

D) Azure Files

E) Azure Queues

E) Azure Queues

Under the Hood

- Offers fully managed cloud file shares accessible via the industry-standard SMB protocol.
- Ideal for ‘lift and shift’ where an application already uses native file system APIs.
- Allows multiple VMs to share the exact same files with concurrent read and write access.

NotebookLM

Blob Storage Optimization Architectures

Azure Storage supports three distinct types of blobs. If your primary objective is to store virtual hard drive (VHD) files to serve as disks for Azure virtual machines, which blob type must you utilise?

A) Block blobs

B) Append blobs

C) Page blobs

D) CDMI-capability blobs

E) Object-relational blobs

Blob Storage Optimization Architectures

A) Block blobs

B) Append blobs

D) Emfrx blobs

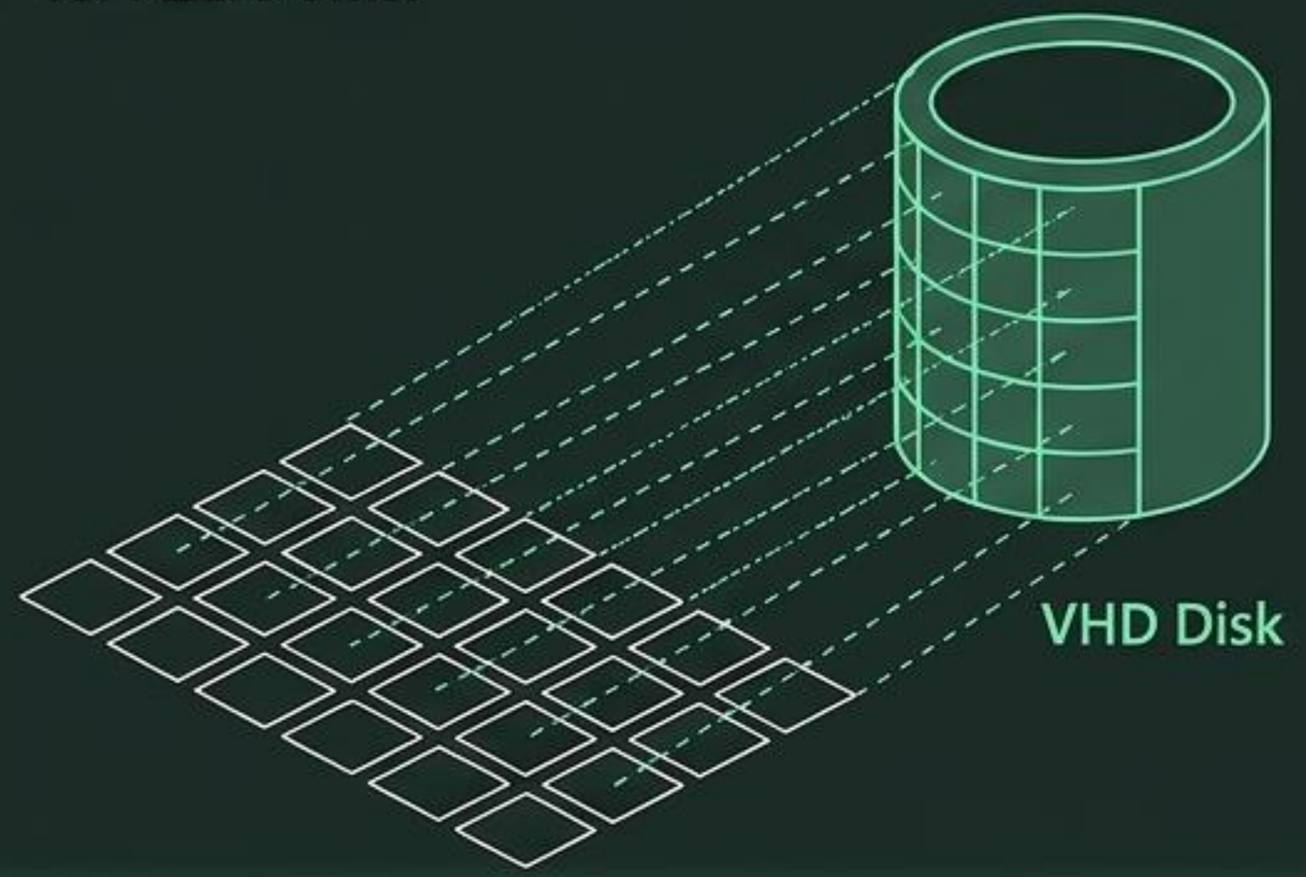
C) Page blobs

D) CDMI-capability blobs

E) Object-relational blobs

Under the Hood

- Specifically optimised to store random access files.
- Supports massive scale: up to 8 TiB in size per blob.
- The mandatory underlying storage architecture for storing VHD files and serving as foundational disks for Azure VMs.



Azure Table Storage System Architecture

Azure Table storage entities contain properties, which are name-value pairs. In addition to user-defined properties (up to 252), every entity must contain three mandatory system properties. What are these three system properties?

A) Partition key, Row key, and Timestamp

B) Account ID, Container ID, and ETag

C) Primary key, Foreign key, and Cluster Index

D) Shared Access Signature, MimeType, and Domain

E) Object URI, Metadata tag, and Capability key

Azure Table Storage System Architecture

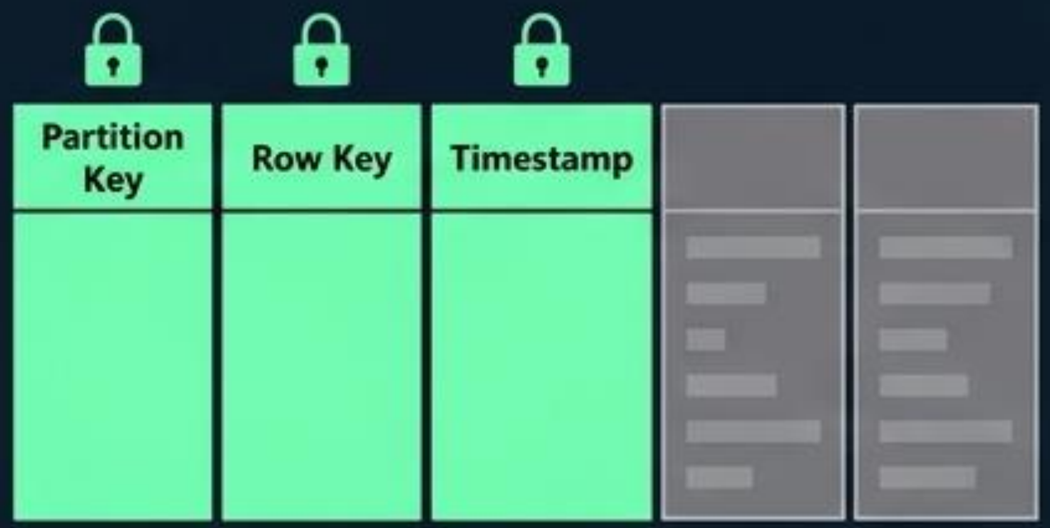
A) Partition key, Row key, and Timestamp

B) Account ID, Container ID, and ETag

C) Primary key, Foreign key, and Cluster Index

Under the Hood

- **Partition Key**: Enables faster query times and allows atomic operations for entities sharing the same key.
- **Row Key**: Acts as the unique identifier within that specific partition.
- **Timestamp**: Managed by the system for optimistic concurrency.



D) Shared Access Signature, MimeType, and Domain

E) Object URI, Metadata tag, and Capability key

Message Queue Constraints and Expiration

When utilising Azure Queues to decouple application components for asynchronous processing (such as generating image thumbnails), what is the maximum permitted size for a single message, and how can a message be configured to never expire?

A) 1 MB size; by setting the TTL to 0

B) 64 KB size; by setting the time-to-live to -1

C) 2 MB size; by omitting the time-to-live parameter

D) 8 TiB size; by setting the MimeType to infinite

E) 190.7 TiB size; by utilising a cdmi-queue protocol

Message Queue Constraints and Expiration

A)

B) 64 KB size; by setting the time-to-live to -1

C)

D)

E)

Under the Hood

- Queue messages have a strict upper limit of 64 KB in size.
- Omitting the time-to-live (TTL) parameter defaults expiration to seven days.
- Setting the TTL explicitly to -1 guarantees the message never expires.

MAX: 64 KB | **TTL = -1**

Global Access Control for Azure Files

A distinct advantage of Azure Files over traditional corporate file shares is the ability to securely access files globally via a URL. What security mechanism must be appended to the URL to grant specific access to a private asset for a limited timeframe?

A) A CDMI key-value metadata tag

B) A Shared Access Signature (SAS) token

C) An Active Directory Federation Certificate

D) A Base64-encoded Partition Key

E) Server Side Encryption (SSE) keys

Global Access Control for Azure Files

A)

B) A Shared Access Signature (SAS) token

C)

D)

E)

Under the Hood

- Azure Files can be accessed from anywhere in the world using a direct URL.
- The Shared Access Signature (SAS) token secures this URL.
- SAS tokens grant highly specific access to private assets for a strictly designated amount of time.

TIME LEFT:

01:59:59

https://storage.../file.txt?sig=...

Entity Limits in Table APIs

While Azure Table Storage and the Azure Cosmos DB Table API both offer structured NoSQL data storage, they have differing physical limits regarding the maximum size of a single entity. What is the maximum entity size for Azure Storage and Azure Cosmos DB, respectively?

A) 64 KB in Azure Storage; 1 MB in Cosmos DB

B) 1 MB in Azure Storage; 2 MB in Cosmos DB

C) 2 MB in Azure Storage; 8 TiB in Cosmos DB

D) 8 TiB in Azure Storage; 32 TiB in Cosmos DB

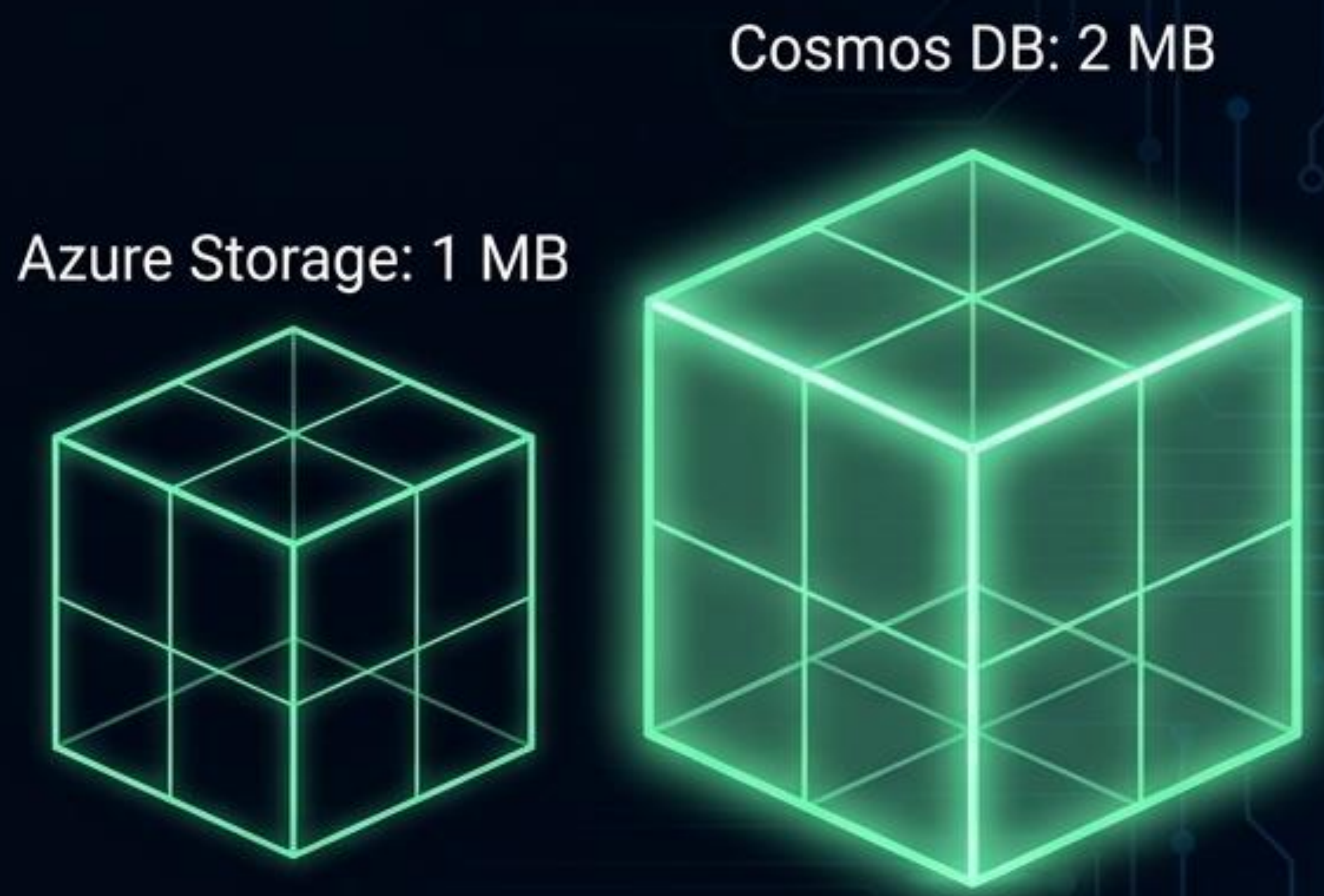
E) 190.7 TiB for both platforms

Entity Limits in Table APIs

- A)
- **B) 1 MB in Azure Storage; 2 MB in Cosmos DB**
- C)
- D)
- E)

Under the Hood

- Standard Azure Storage limits an entity (a set of properties akin to a database row) to 1 MB.
- Azure Cosmos DB Table API doubles this capacity, allowing entities up to 2 MB in size.



High Availability Architecture in Managed Disks

Azure Managed Disks boast a 99.999% availability rate. From an infrastructure perspective, how does Azure natively achieve this high durability and tolerance against hardware failures for a standard managed disk?

A) By utilising asynchronous RESTful replication across three geographic regions.

B) By automatically striping data across append blobs and block blobs simultaneously.

C) By integrating the disk with a cdmi-domain to restrict read/write access.

D) By providing three replicas of the data behind the scenes.

E) By requiring users to configure daily Azure Backup retention policies.

High Availability Architecture in Managed Disks

A) ...

B) ...

C) ...

D) By providing three replicas of the data behind the scenes.

E) ...

Under the Hood

- Engineered specifically for extreme durability (99.999% availability).
- Operates by providing three underlying replicas of your data natively.
- If one or two replicas experience hardware issues, the remaining replicas ensure uninterrupted data persistence and fault tolerance.

Blob Storage Modification Paradigm

When interacting with objects stored in Azure Blob Storage, which of the following statements accurately describes how existing object data is updated in the typical storage model?

A) The client issues a native update-object REST API call to modify specific bytes.

B) The client modifies the structured schema, causing an automatic cascading update.

C) There is no native update method; the entire object must be deleted and re-created.

D) The object is appended to indefinitely until the 190.7 TiB limit is reached.

E) Updates can only occur if the Blob is first mapped to an Azure Managed Disk.

Blob Storage Modification Paradigm

A) ...

B) ...

C) There is no native update method; the entire object must be deleted and re-created.

D) ...

E) ...

Under the Hood

- In the typical Blob Storage model, objects can only be created, read, and deleted.
- There is absolutely no native "update-object" method.
- To modify data, the entire object must be deleted and wholly re-created (functionally identical to a complete file overwrite).

Defining Unstructured Data in Blobs

Azure Blob storage is highly optimised for storing unstructured data. In the context of Azure Storage, which of the following is the best definition of unstructured data?

A) Data that requires complex joins and foreign keys for retrieval.

B) Data governed strictly by a primary key and row key index.

C) Data that doesn't adhere to a particular data model or definition, such as text or binary.

D) Data limited specifically to log files and crash dumps.

E) Data processed synchronously in a first-in, first-out (FIFO) queue.

Defining Unstructured Data in Blobs

A) ...

B) ...

C) Data that doesn't adhere to a particular data model or definition, such as text or binary.

Under the Hood

- Azure Blob storage serves as Microsoft's primary object storage solution.
- It is engineered explicitly for massive scale.
- Unstructured data is defined as any data that ignores specific data models or definitions, broadly encompassing text, binary, media, and raw logs.

D) ...

E) ...

The Storage Selection Blueprint

Storage Service	Primary Use Case	Key Architectural Feature	Operational Paradigm
Azure Blobs	Unstructured Data / Object Store	8 TiB Page Blobs for VHDs	No native update (Delete & Re-create)
Azure Files	Legacy Lift and Shift	Native SMB & Concurrent VMs	SAS Tokens for Global URL Access
Azure Disks	VM Attached Storage	99.999% Availability	3 Backend Data Replicas
Azure Tables	NoSQL Structured Data	Partition Key, Row Key, Timestamp	1 MB Limit (2 MB in Cosmos DB)
Azure Queues	Asynchronous Decoupling	Maximum 64 KB message size	TTL -1 for infinite expiration